

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

Course Code: 23CS002PC304

Course Title: AI Assisted Coding

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Lab Objectives

- Understand advanced string algorithms
- Implement Z-algorithm for pattern matching
- Analyze prefix matching techniques
- Improve efficiency in string processing

Task 1: Z-Algorithm Implementation

Python Code

```
def build_z_array(s):  
    n = len(s)  
    Z = [0] * n  
    l = r = 0  
  
    for i in range(1, n):  
        if i <= r:  
            Z[i] = min(r - i + 1, Z[i - l])
```

```

        while i + Z[i] < n and s[Z[i]] == s[i + Z[i]]:
            Z[i] += 1

        if i + Z[i] - 1 > r:
            l, r = i, i + Z[i] - 1

    return Z

```

Task 2: Pattern Matching using Z-Algorithm

Python Code

```

def find_pattern(text, pattern):
    concat = pattern + "$" + text
    Z = build_z_array(concat)
    m = len(pattern)

    for i in range(len(Z)):
        if Z[i] == m:
            print("Pattern found at index", i - m - 1)

find_pattern("abxabcabcaby", "abcaby")

```

Task 3: Explanation

- Z-array stores longest substring starting from index i
- Helps match pattern with all suffixes efficiently
- Time complexity: $O(n + m)$

Conclusion

Z-algorithm provides efficient pattern matching.

AI tools help implement complex algorithms quickly and accurately.